



BME

BUDAPEST UNIVERSITY
OF TECHNOLOGY AND ECONOMICS

Pharmacy

Management System

Desktop Application

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INTRODUCTION

01

Fragmented Operations

Stock, sales, prescriptions, and returns managed separately — no unified interface.

02

Traceability Risk

Manual processes leave gaps in audit trails; no accountability on user actions.

03

Expiry & Stock Risk

Medicine batches expire unnoticed; low-stock alerts prevent stockouts.

MAIN GOAL

One unified desktop application to manage:

- Products & Batches
- Point of Sale
- Prescriptions
- Returns & Refunds
- Audit Logs

Role-based: Admin & Pharmacist

SYSTEM OVERVIEW



Dashboard

Metrics, alerts



Products

Catalog & barcodes



Batch Mgmt

Stock & expiry



Point of Sale

Sales & cart



Prescriptions

Rx records



Sales History

Receipts & review



Returns

Refund tracking



Audit Logs

Action trail



Settings

User management

Workflow: Products → Batches → POS → Prescriptions/Receipts → Returns → Audit Logs

TECHNOLOGY STACK

01

Python 3

Application language

Cross-platform, readable, large ecosystem

02

PyQt6

Desktop GUI framework

Native widgets, event-driven, role-based views

03

SQLite

Local database engine

Embedded, zero-config, file-based storage

Modular page structure

Role-based logic

Local SQLite DB

Desktop GUI (PyQt6)

INVENTORY MANAGEMENT

PRODUCTS

- Name, dosage form, barcode
- Category & pricing
- Prescription flag (Rx)
- Stock threshold alert

The screenshot shows the 'Products' management page. On the left is a sidebar with navigation links: Dashboard, Batch Management, Point of Sale, Product (highlighted), Prescriptions, Sales History, Returned Items, Audit Logs, and Settings. The main header includes the user 'Khalis Dawa Administrator Mode', the date 'Wednesday, June 03, 2026', and a 'Logout' button. The page title is 'Products' with a subtitle 'View and manage product records' and an '+ Add Product' button. A search bar 'Search products...' and a category filter 'All Categories' are present. The table lists products with columns: NAME, FORM, BARCODE, PRICE, MIN STOCK, CATEGORY, RX, AVAILABLE, and ACTIONS. The data rows are:

NAME	FORM	BARCODE	PRICE	MIN STOCK	CATEGORY	RX	AVAILABLE	ACTIONS
Ambroxol Syrup	Syrup	859465572844	640 Ft	20	Respiratory	No	0 units	Edit Delete
Amlodipine	Tablet	131861401307	920 Ft	8	Cardiac	No	0 units	Edit Delete
Amoxicillin	Capsule	166168184048	1200 Ft	10	Antibiotic	Yes	5 units	Edit Delete
Atenolol	Tablet	1655024408522	780 Ft	8	Cardiac	No	0 units	Edit Delete
Augmentin	Tablet	680970451386	2450 Ft	6	Antibiotic	Yes	0 units	Edit Delete

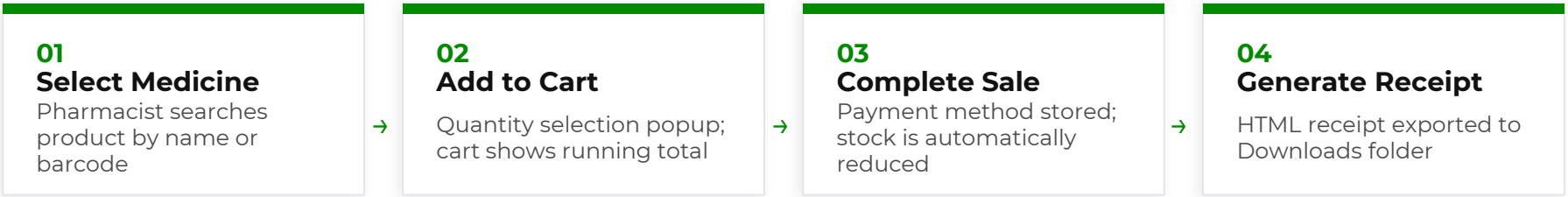
BATCH MANAGEMENT

- Stock quantity per batch
- Expiry date tracking
- Dashboard expiry warnings
- Low stock dashboard alerts

The screenshot shows the 'Batch-based Inventory Management' page. The sidebar is identical to the previous screenshot. The main header shows the same user and date. The page title is 'Batch-based Inventory Management' with a subtitle 'Product Batch Manager' and an '+ Add Batch' button. A search bar 'Search medicines...' and a category filter 'All Categories' are present, along with a checkbox for 'Expired Batches'. The table lists batches with columns: MEDICINE, CATEGORY, BATCH, STOCK, UNIT PRICE, EXPIRY, and ACTIONS. The data rows are:

MEDICINE	CATEGORY	BATCH	STOCK	UNIT PRICE	EXPIRY	ACTIONS
Ambroxol Syrup	Respiratory	B024-A	10 units	640.0 Ft	Expired 2026-04-15	Edit Delete
Amlodipine	Cardiac	B020-A	5 units	920.0 Ft	Expired 2025-06-15	Edit Delete
Amoxicillin	Antibiotic	B004-A	8 units	1200.0 Ft	Good 2027-06-15	Edit Delete
Amoxicillin	Antibiotic	B004-B	5 units	1200.0 Ft	Good 2027-10-20	Edit Delete
Atenolol	Cardiac	B031-B	7 units	780.0 Ft	Expiring soon 2026-5-30	Edit Delete
Atenolol	Cardiac	B031-A	4 units	780.0 Ft	Expired	Edit Delete

SALES WORKFLOW



Khalis Dawa

Administrator Mode

Dashboard

Batch Management

Point of Sale

Product

Prescriptions

Sales History

Returned Items

Audit Logs

Settings

Logout

Wednesday, June 03, 2026

Point of Sale

Search products, add medicines to cart, and checkout

Search medicines by name or barcode

All Categories

740 Ft 666 Ft -10% Add

540 Ft 486 Ft -10% Add

6329 Ft 5696.1 Ft -10% Add

Hydrocortisone

Dermatology

Available: 12

680 Ft 612 Ft -10% Add

Losartan

Cardiac

Available: 12

9100 Ft 990 Ft -10% Add

Mounjaro

Diabetes

Available: 12

1100 Ft 990 Ft -10% Add

Multivitamin

Supplement

Available: 12

6075 Ft 5467.5 Ft -10% Add

Loratadine

Allergy

Available: 12

540 Ft 486 Ft -10% Add

Metronidazole

Antibiotic

Available: 11

720 Ft 648 Ft -10% Add

Nasal Spray

ENT

Available: 11

720 Ft 648 Ft -10% Add

Batch Picking Instructions

Pick items from these batches

Insulin Regular x3

Take 2 from batch B026-A (expires 2026-06-15)

Take 1 from batch B026-B (expires 2026-10-20)

Multivitamin x5

Take 5 from batch B019-A (expires 2026-06-15)

Total: 17415 Ft

Cancel Confirm Checkout

Cart

Insulin Regular x3 = 11340 Ft

Multivitamin x5 = 6075 Ft

Total: 17415 Ft

Empty Cart Checkout

Invoice

Khalis Dawa

Pharmacy Receipt

#000024

Date: 2026-06-03 00:57

Payment Method: Cash

No.	Item	Quantity	Unit Price	Subtotal
1	Insulin Regular	3	3780 Ft	11340 Ft
2	Multivitamin	5	1215 Ft	6075 Ft
TOTAL				17415 Ft

Thank you for shopping with Khalis Dawa.

Khalis Dawa

kunigunda uja 26, 1037, BUDAPEST

dummymail@khalisdawa.com

PRESCRIPTION HANDLING

HOW IT WORKS

- 1 POS detects Rx flag on selected medicine
- 2 Customer name & SSN collected before sale
- 3 Non-Rx purchases skip this step entirely
- 4 All Rx records stored in Prescriptions page
- 5 Prescription report exportable per entry

Point of Sale
Search products, add medicines to cart, and checkout

Search medicines by name or barcode All Categories

Ambroxol Syrup
Respiratory
Available: 0
640 Ft

Atenolol
Cardiac
Available: 0
780 Ft

Blood Glucose Strips
Diagnostic
Available: 17
2100 Ft 1890 Ft -10%

Customer Details
SSN and name are required because the cart contains prescription products.

Customer Name:

Customer SSN:

Amoxicillin
Antibiotic
Available: 5
1200 Ft

Azithromycin
Antibiotic
Available: 10
1650 Ft 1485 Ft -10%

Cetirizine Syrup
Allergy
Available: 12
470 Ft 423 Ft -10%

Calpol
Children
Available: 4
780 Ft

Cart
Azithromycin x2 = 2970 Ft

Total: 2970 Ft

Khalis Dawa
Administrator Mode

Wednesday, June 03, 2026

admin Administrator Mode

Prescriptions
Track sensitive medicine sales

Search by SSN, customer, or medicine...

SSN	CUSTOMER	MEDICINE	AMOUNT	DATE	REPORT
64376688	Anthony Blake	Azithromycin	2	2026-06-03 09:13	Report
647596756	Peter Brooker	Insulin Regular	3	2026-06-03 08:57	Report
564796756	Peter Brooker	Insulin Regular	3	2026-06-03 08:49	Report
374916393	Ivan Brooker	Metronidazole	1	2026-06-03 08:37	Report
374916393	Ivan Brooker	Insulin Regular	1	2026-06-03 08:37	Report
546897536	James Carcione	Azithromycin	4	2026-06-03 08:35	Report
777888999	jk	Amoxicillin	5	2026-05-31 18:23	Report
444555666	jk	Amoxicillin	2	2026-05-31 18:14	Report
398447349	Muhammad Hameez Khan	Amoxicillin	3	2026-05-31 17:17	Report

RETURNS & AUDIT TRAIL

RETURNED ITEMS

- Refund amount calculated automatically
- Expiry check before accepting return
- Stock restored to original batch
- Return report exported as HTML

AUDIT LOGS

- Logs all meaningful admin & pharmacist actions
- Timestamps recorded per event
- Improves operational accountability
- Non-editable — read-only view

The screenshot shows the 'Returned Items' page in Administrator Mode. The left sidebar contains navigation links: Dashboard, Batch Management, Point of Sale, Product, Prescriptions, Sales History, Returned Items (highlighted), Audit Logs, and Settings. The main content area has a search bar and a table of returned items. The table columns are Receipt ID, Items, Date Returned, Refund, Source Sale, Report, and Actions. The 'Report' column contains 'Download' buttons, and the 'Actions' column contains 'Delete' buttons.

RECEIPT ID	ITEMS	DATE RETURNED	REFUND	SOURCE SALE	REPORT	ACTIONS
#00006	3	2026-06-03 01:22	5328 Ft	#00023	Download	Delete
#00005	1	2026-06-03 01:22	405 Ft	#00025	Download	Delete
#00004	5	2026-05-31 18:23	6000 Ft	#00020	Download	Delete
#00003	2	2026-05-31 18:14	2400 Ft	#00019	Download	Delete
#00002	1	2026-05-31 17:49	1200 Ft	#00018	Download	Delete
#00001	2	2026-05-31 16:21	1560 Ft	#00017	Download	Delete

The screenshot shows the 'Audit Logs' page in Pharmacist Mode. The left sidebar contains navigation links: Dashboard, Batch Management, Point of Sale, Product, Prescriptions, Sales History, Returned Items, Audit Logs (highlighted), and Settings. The main content area has a search bar and a table of audit logs. The table columns are Username, Role, Description, and Date.

USERNAME	ROLE	DESCRIPTION	DATE
pharmasuer1	pharmacist	Sold Cymbalta x2	2026-06-03 01:25
pharmasuer1	pharmacist	Signed in	2026-06-03 01:24
pharmasuer1	pharmacist	Created pharmacist account	2026-06-03 01:24
admin	admin	Carried out item return for Metformin x2 from #00023	2026-06-03 01:22
admin	admin	Carried out item return for Insulin Regular x1 from #00023	2026-06-03 01:22
admin	admin	Carried out item return for Paracetamol x1 from #00023	2026-06-03 01:22

DASHBOARD OVERVIEW

Medicines

Total products tracked

Sales

Completed transactions

Returns

Processed refunds

Alerts

Low stock + expiry

Khalis Dawa

Pharmacist Mode

Dashboard

Batch Management

Point of Sale

Product

Prescriptions

Sales History

Returned Items

Audit Logs

Logout

Wednesday, June 03, 2026

pharmauser1 Pharmacist Mode

Dashboard

Home Page

Medicines

48

Sales

21

\$

Returned Items

14

R

Expiration Alert

11

!

Low Stock

17

L

Revenue

109199 Ft

Ft

Low Stock Products

PRODUCT	AVAILABLE	MIN REQUIRED
Ambroxol Syrup	0	20
Amlodipine	0	8
Atenolol	0	8
Augmentin	0	6
jfhvjhg	0	20

Batches expiring within 30 days

PRODUCT	BATCH	EXPIRY DATE
Paracetamol	B003-Z	2026-06-05
Zepbound	ZEP-001	2026-06-15
Paracetamol	B001-A	2026-06-15
Calpol	B003-A	2026-06-15
Cetirizine	B005-A	2026-06-15

IMPLEMENTATION SUMMARY

- Full desktop app (Python + PyQt6 + SQLite)
- Role-based login system (Admin / Pharmacist)
- Product catalog with barcode & category support
- Batch management with expiry & low-stock alerts
- Point of Sale with cart & receipt generation
- Prescription tracking with customer SSN storage
- Returns management with batch stock restoration
- Audit log for all user actions
- Dashboard with live pharmacy metrics

THESIS SEMESTER PLAN & FUTURE WORK

THESIS SEMESTER PLAN

Packaged Windows installer

Automated test suite (unit + integration)

Database backup & export tools

Richer reporting & analytics charts

Thesis document & final review

FUTURE IMPROVEMENTS

- Dark theme support
- Multi-language (EN / HU)
- Data export to CSV / Excel
- Enhanced analytics & charts
- Improved UI reliability
- Cloud sync option

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THANK YOU